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Roll No. | | | | | | | | | | | | | **Total No. of Pages : 03**

Total No. of Questions : 09

B.Tech. (CE) (Sem.–6)

PRE-STRESSED STRUCTURES

Subject Code : PECE-603E-18

M.Code : 79404

Date of Examination : 19-12-2022

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

SECTION-A

1. Write briefly :

a) Write down difference between pre-tensioning and post tensioning.

b) Write down the assumptions in the design of pre-stressed concrete members.

c) What is the role of high strength steel in pre-stressed concrete?

d) Define pre-stressing.

e) What are the different applications of pre-stressed concrete?

f) Write down the requirement of different materials for pre-stressed concrete members.

g) Explain Freyssinet system of pre-stressing.

h) Write down a short note on P.S.C Monowire system.

i) What is the need of pre-stressing?

j) Explain the MagnelBlaton system of pre-stressing.

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SECTION-B

2. A pre-stressed concrete beam 250mm wide and 375mm deep is pre-stressed by concentrically placed tendon. The span of the beam is 8m and the beam has to support an imposed load of 4.25kN/m. Find the pre-stressing force necessary so that the tension is just avoided at the soft of mid span. If tendon is provided at an eccentricity of 65mm find the pre-stressing force necessary so that tension is just avoided at the sofit of mid-section.

3. A pre-tensioned beam of rectangular section 80mm wide and 120mm deep is to be designed to support concentric loads of 4kN each at one third span points over an effective span of 3m. The permissible stresses in concrete are limited to zero and $1.4 N/mm^2$ in tension at transfer and working load respectively. If 3 mm dia wires initially stressed to $1400 N/mm^2$ are used, find the number of wires required and the eccentricity of the pre-stressing force assuming 20 percent loss in pre-stress.

4. A pre-stressed concrete pile is 300mm* 3 00mm in section and is provided with 40 wires of 3mm dia distributed uniformly over the section. Initially the wires are tensioned in the pre-stressing beds with a total pull of 450kN. Determine the final stress in concrete and the percentage loss of stress in the wires. Take $Es = 2.08 \times 10^5 N/mm^2$, $Ec = 3.20 \times 10^4 N/mm^2$, Creep shortening = 32×10^{-6} mm/mm per N/mm^2 of stress, Total shrinkage strain = 200×10^{-6}, Relaxation loss of stress in steel = 4.5% of initial stress.

5. Write a detailed note on different types of losses of pre-stress.

6. Write down a detailed note on concept of load balancing.

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SECTION-C

7. A pre-stressed concrete beam of rectangular section is 125mm wide and 250mm deep and has a span of 6m. The beam is provided with a straight tendon at a uniform eccentricity of 40mm, the pre-stressing force being 190kN. Find the deflection at the centre.

   a) Under the action of pre-stress and dead load of the beam

   b) Under the action of pre-stress, dead load and live load of 3.75kN/m and including the effect of creep and shrinkage, taking the creep coefficient as 1.75

   Compare these deflections with the permissible limits.

8. Design a simply supported pre-stressed concrete slab to the following conditions: Span of the slab: 12m, Safe stress in concrete = 14N/mm, Safe stress in steel = 900N/mm^2, Superimposed load = 20kN/mm^2, Weight of PSC = 24kN/mm^2

9. Design a pre-stressed concrete beam to the following requirements:

   Span = 15m, Superimposed load = 34kN/m, Cube strength of concrete at 28 days = 35N/mm^2, Safe stress in concrete at transfer of pre-stress = f_r = 0.5f_{Ck}, Safe stress in concrete due to final pre-stress = fc = 0.4fck, Total loss of pre-stress = 20%, Allowable tensile stress in concrete = 0.219f_{ck}, Ultimate stress in steel = 1500N/mm^2, Safe stress in steel = 60% of ultimate stress.

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